

Final Review of Related Literature (RRL)

Final review of related literature refers to the process of evaluating and analyzing literature that will serve as the basis for the development of the SCHOOLCARE+ Web-based School Clinic Information System. This review will ensure that the gathered literature is related to the problem and will provide the best possible basis for the proposed system.

The related literature has been analyzed according to the different elements that make up the proposed system such as electronic health record management, medical inventory management, scheduling appointments, automated reporting and data security. Studies that relate to the Electronic Health Record (EHR) system stressed the need for digital systems in order to improve the accessibility and organization of the healthcare information. Literature that was written about Health Information Technology (HIT) supported the use of digital technologies in order to minimize mistakes in documentation and make healthcare services efficient.

Studies regarding inventory management system stressed the significance of monitoring the availability of medicine, their expiration date and movements within the organization. Lastly, studies regarding Role-Based Access Control (RBAC) stressed the significance of access control management in order to protect the confidentiality of student medical information.

Lastly, it was noted that most of the existing solutions have concentrated more on healthcare-related tasks such as record management or inventory control, while less attention has been paid to creating a comprehensive solution that encompasses all these tasks in a single school

clinic management system. Thus, the research gap has encouraged the creation of SCHOOLCARE+.

Final Review of Conceptual Framework

The conceptual framework of SCHOOLCARE+ is done through the Input-Process-Output (IPO) Model, which presents the conversion of the identified problems and requirements into an effective school clinic information system.

Input

The inputs are the current problems and requirements taken from the school clinics, which include:

- Manual maintenance of student medical record
- Difficulty in accessing student's health information
- Inefficient management of medicines and medical supplies
- Lack of proper scheduling of appointments
- Challenges in producing health reports
- Access control to prevent access to confidential information

These inputs will be used as basis in determining the features needed in the system.

Process

The processes are the activities that should be undertaken in designing and developing the SCHOOLCARE+ system, such as:

- Requirements and data gathering
- System and architecture design
- Database design and development
- Front end and back end development

- Incorporation of modules for health records, inventory, scheduling, and report generation
- Security through implementation of authentication and Role-Based Access Control (RBAC)

By undertaking the mentioned processes, the identified clinic problems will be transformed into digital solutions.

Final Review of Research Methodology

The review of the research methodology is important to ensure that the research methods applied in the development of the proposed SCHOOLCARE+ system are appropriate and in line with the study objectives. The researchers employed the system development method of designing, developing, testing, and evaluating the proposed web-based school clinic information system.

Methodology Stages Involved

Requirements Analysis

The researchers analyzed the problems existing in the school clinics, especially with regards to management of students' medical records, inventory control, appointment scheduling, and report generation. Requirements obtained from the analysis became the basis for the definition of system functionalities.

System Design

The researchers created system architecture, database, user interface, and system modules from the identified requirements. System design was aimed at ensuring that the system had proper organization and user-friendliness.

System Development

SCHOOLCARE+ system was developed employing modern web technologies. In the development process, front-end interface was developed using Vite React, HTML, CSS, TypeScript, and Tailwind CSS, while back-end system was developed using Node.js, Express.js, REST API, and MongoDB.

Testing and Evaluation

The developed system was tested to ascertain its functionality, usability, reliability, and efficiency. Testing covered the following modules:

Student Medical Record Management

- Consultation Tracking
- Medicine Inventory Management
- Appointment Scheduling
- Automatic Report Generation
- User Authentication and Access Control
- Feasibility Analysis

The following feasibility factors were considered by the researchers to determine the viability of the proposed system:

Technical Feasibility – assesses the possibility of utilizing existing technologies and other resources in developing the system.

Economic Feasibility – examines the practicality of using free software and minimizing manual processes.

Operational Feasibility – considers the efficiency of implementing the system and meeting user demands.

Legal Feasibility – ensures that the system meets the standards of data privacy laws, including the Data Privacy Act of 2012 (RA 10173).

Schedule Feasibility – ensures that the system is implementable within the specified schedule.

The final review of methodologies proves that the selected development process is appropriate for developing the SCHOOLCARE+.